TEXORA · LECTURE SHEET

Optical Properties

Light in an anisotropic fibre: two refractive indices instead of one, birefringence as the fingerprint of molecular orientation, retardation and interference colours under the polarising microscope, and the surface optics of lustre that the eye reads as quality.

TE-3209 · TEXTILE PHYSICS · TOPIC 6 · LEVEL 3 TERM 2

1. Two refractive indices

A fibre's chains run (more or less) along its axis, so light polarised parallel to the axis experiences a different refractive index $n_{\parallel}$ than light polarised across it, $n_{\perp}$. The difference is the **birefringence**:

$$\Delta n = n_{\parallel} - n_{\perp}$$

FIBRE	N∥	N⊥	ΔN — AND WHAT IT REPORTS
Polyester (drawn)	≈ 1.71	≈ 1.54	≈ 0.16 — very high orientation
Nylon 6.6	≈ 1.58	≈ 1.52	≈ 0.06
Cotton	≈ 1.578	≈ 1.532	≈ 0.046 — grown-in chain alignment (microfibril spiral lowers the apparent value)
Viscose	≈ 1.54	≈ 1.52	≈ 0.02 — modest orientation, process-variable
Wool	≈ 1.556	≈ 1.546	≈ 0.010 — near-isotropic chains arrangement

Δn is therefore a **non-destructive orientation meter**: draw a filament more and Δn climbs with tenacity; heat-relax it and both fall. It is the routine research gauge for spinning and drawing quality, and the identity card used in fibre identification (with density and melting behaviour) for unknown filaments.

2. Retardation and interference colours

Between crossed polars, the two rays through a fibre travel at different speeds and emerge with a path difference, the **retardation**:

$$\Gamma = t \times \Delta n \quad (t = \text{fibre thickness})$$

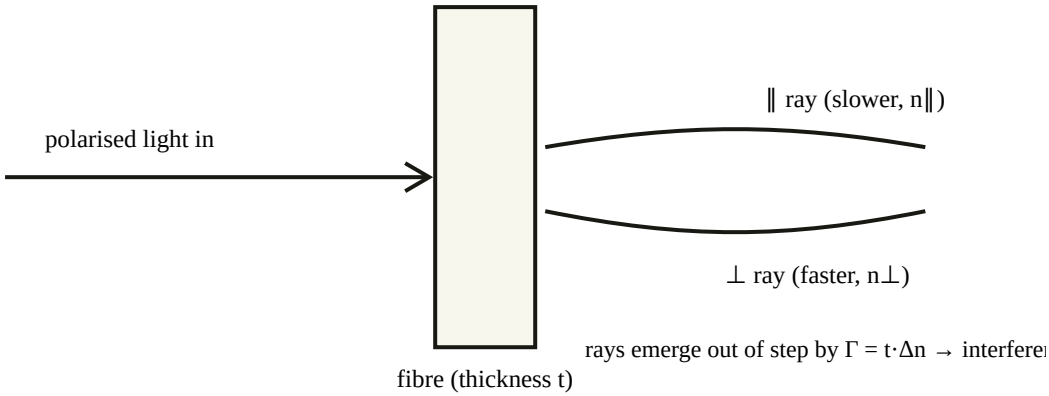


Figure 1 — Polarised light splitting into two rays inside a birefringent fibre: the parallel ray is slowed more, and the pair recombine at the analyser with retardation Γ, producing the Michel-Lévy interference colours.

WORKED EXAMPLE — RETARDATION AND COLOUR ORDER

A cotton fibre 20 µm thick with Δn = 0.046: Γ = 20,000 nm × 0.046 = **920 nm** — a second-order interference colour (blue-green region of the Michel-Lévy chart). A drawn polyester filament of the same thickness with Δn = 0.16: Γ = **3200 nm** — high-order, washing out toward pearly white. The microscopist reads Γ with a compensator, divides by measured thickness, and has Δn — and with it, orientation — in minutes. The same physics makes stress patterns visible in transparent plastics (photoelasticity).

3. Lustre — the surface’s contribution

Lustre is the ratio and geometry of specular to diffuse reflection: smooth, round, aligned filaments mirror light coherently (bright silk, filament yarns); rough, twisted, matt surfaces scatter it. Industry tunes it deliberately — **delustrant** TiO_2 particles inside synthetic fibres scatter light internally (full-dull, semi-dull, bright grades), trilobal cross-sections break reflections into sparkle, yarn twist and fabric weave spread or concentrate the sheen, and calendering presses a temporary polish onto cotton. Perceived quality — the “silk-like” polyester, the “rich” matt wool — is applied optics, set by Δn-independent surface choices.

Two smaller optical properties earn their keep: **dichroism** — dyed fibres absorb differently for the two polarisations when dye molecules align with chains, a research probe of dye location and orientation — and **fluorescence**: optical brightening agents absorb UV and re-emit blue, compensating yellowness (a loan from the spectrum that washing gradually calls in).

4. Key points

- Anisotropic fibres carry two indices; $\Delta n = n_{\parallel} - n_{\perp}$ measures molecular orientation (PET 0.16 → wool 0.010).
- Retardation $\Gamma = t\Delta n$ produces interference colours: 920 nm for the cotton example — Δn is measurable at the microscope with a compensator.
- Drawing raises Δn, tenacity together; relaxation lowers both — Δn is the process engineer's orientation gauge.
- Lustre is surface optics: delustrants, cross-section and twist tune it independently of the polymer.
- Dichroism probes dye orientation; fluorescence (brighteners) borrows UV to fake whiteness.

5. Practice

Answer in your own words; brief model answers follow.

1. A filament 15 µm thick shows retardation 900 nm. Find its birefringence and guess the fibre family.
2. Two PET filaments, A drawn 4.5× and B drawn 2.5×, have Δn 0.15 and 0.08. Predict their relative tenacity and dye uptake, justifying by orientation.
3. Why does cotton's apparent Δn (0.046) fall below what its crystalline chain alignment alone would give?
4. A "full-dull" and a "bright" polyester fabric differ in appearance though woven identically from the same polymer. Explain the mechanism and name the additive.
5. An unknown white filament shows $\Delta n \approx 0.06$, melts near 260 °C and has density 1.14 g/cm³. Identify it and name the two confirmatory tests implied by the numbers given.
6. Why do interference colours fade to white at very high retardation, and what does the microscopist do about it when measuring thick, highly drawn fibres?

Answer key

1. $\Delta n = \Gamma/t = 900/15{,}000 = 0.06$ — nylon territory (compare the table: PET would be ~0.16, cotton 0.046).
2. A (higher Δn = more orientation): higher tenacity, lower breaking extension, lower dye uptake — chains aligned to the axis carry load directly, and the tighter structure admits less dye. B dyes deeper but is the weaker filament.
3. Cotton's chains lie in microfibrils spiralling around the fibre axis (fibrillar angle 20–30°): the microscope averages the tilted contributions, so measured Δn is the axial projection of a higher local alignment — geometry dilutes the reading.

4. Titanium dioxide delustrant particles dispersed in the full-dull polymer scatter light inside the filament, destroying the clean surface reflection; the bright grade omits it. Same polymer, same weave — different internal optics.
5. Nylon 6.6: Δn 0.06 ✓, density 1.14 ✓, melting ≈ 255–260 °C ✓. Confirmatory tests: the melting point determination and the density (gradient column) — optical and thermal evidence already triangulate it.
6. At high Γ many wavelengths interfere across the visible band simultaneously; the summed colour washes toward white (high-order white). The microscopist measures individual fibres at thin edges, uses a compensator to bring the order down, or sections the sample — never trusts a colour name alone at high order.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib